AgroLD: a Knowledge Graph for the Plant Sciences

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Abstract
The AgroLD Knowledge Graph is a semantic framework designed to integrate and explore data relevant to plant sciences, particularly focused on plant genomics. AgroLD contains around 900M triples created by combining more than 100 datasets from 15 data sources. Our objective is to offer a domain-specific knowledge platform to answer complex biological and plant sciences questions related to the implication of genes in, for instance, plant disease resistance or adaptative responses to climate change. In this poster, we present some results which currently focused on genomics, genetics, and trait associations.

Keywords
Knowledge Graph, Linked Data, FAIR data, Plant Sciences, Bioinformatics

1. The AgroLD Knowledge Graph

1.1. Overview

AgroLD is built incrementally spanning vast aspects of plant molecular interactions. The current phase covers information on genes, proteins, predictions of homologous genes, metabolic pathways, plant trait associations, and genetic studies. At this stage, we have integrated data from several resources such as Ensembl plants, TAIR, and Gene Ontology Annotation. The choice of these sources has been guided by the biological community, as they are widely used and have a strong impact on the user's confidence. We have also integrated resources developed by the local SouthGreen platform[1] such as TropGeneDB, a tropical plant genetics database, Rice Genome Hub, a rice genomics database, GreenPhylDB, a comparative genomics database for tropical plants, OryzaTagLine, a rice phenotype database and SniPlay, a rice genomic variation database. These resources bring together experimental data produced by research groups in Montpellier and the South of France. The online documentation provides an overview of the integrated data sources[2]. The conceptual framework of AgroLD is based on well-established ontologies in the plant field such as Gene Ontology, Plant Ontology, or Plant Trait Ontology. Furthermore, we developed a dedicated schema [3] that creates links between the imported ontologies and introduces new classes and properties. The online documentation [4] shows the complete list of the used ontologies. The majority of these ontologies are hosted by the OBO Foundry project.

1.2. Statistics

As of today, AgroLD contains more than 900 Millions triples resulting of the integration of roughly 100 datasets gathered in 33 named graphs. Table 1 gives an overview of available resources and tools. All datasets are available in Zenodo under the Creative Commons Attribution 4.0 International license

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[1]https://www.southgreen.fr/
[2]http://www.agrold.org/documentation.jsp
[3]https://github.com/SouthGreenPlatform/AgroLD
[4]http://www.agrold.org/documentation.jsp

Table 1
Links to AgroLD resources and tools

Name of resource or tool and description	URL
Data	
AgroLD datasets	https://doi.org/10.5281/zenodo.4694518
List of graphs	http://www.agrold.org/documentation.jsp
List of ontologies	http://www.agrold.org/documentation.jsp
AgroLD vocabulary	https://github.com/SouthGreenPlatform/AgroLD_ETL/tree/master/model
AgroLD SPARQL Endpoint	http://agrold.southgreen.fr
Example queries	http://www.agrold.org/sparqleditor.jsp
Tools	
Web application	https://github.com/SouthGreenPlatform/AgroLD_webapp
RDF conversion pipelines (GFF2RDF, GAF2RDF, VCF2RDF, Datasets)	https://github.com/SouthGreenPlatform/AgroLD_ETL
Publications	
Original paper	https://doi.org/10.1371/journal.pone.0198270
Resource paper	https://doi.org/10.1007/978-3-030-88361-429

(CC-BY 4.0). Each resource can contain several datasets, for instance, one dataset per species or per data type. Combining all ontologies and datasets imported, the AgroLD graph gather 383 classes and 793 properties. Among the pipelines developed to lift the datasets, we focused also on connecting our datasets with others. The property *rdfs:seeAlso* reaches the total number of almost 80 million of outbound links making the AgroLD graph correctly linked with other datasets in the LOD. Besides, we paid attention to increasing the number of semantic annotations with imported ontologies, which increased the number of links between datasets making the overall graph denser. We created more than 14 million semantic links linking entities to ontological classes. Finally, our data linking strategy allowed us to create around 160,000 *owl:sameAs* links between entities

Declaration on Generative AI

The authors have not employed any Generative AI tools.